

Object Oriented Programming using C++
Department of CSE
Punjabi University Patiala
MST -2

Time 60 Min

MM:15

Section-A (Do All)

- Q.1 What do you mean by Abstract class?.....1
- Q.2 What is the difference between Early Binding and Late Binding ?.....2
- Q.3 Name the two types of Templates in C++.....1
- Q.4 What is a NULL Pointer?.....1

Section-B (Do any Two) (5 mark each)

- Q.6 What is Inheritance? Explain the Ambiguity in Multiple Inheritance and How it can be resolved?.....5
- Q.7 Write a Program to Copy the Contents of one File to another File.....5
- Q.8 Explain Pointer to Objects with the help of a Program.....5

*Note: Do Mention your Groups and Full roll Nos

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, PUNJABI UNIVERSITY, PATIALA
MST-II, OOPS USING C++ (CPE-202), MAX. MARKS 15, TIME 1 HOUR

1. What do you understand by:-

- ☒ a) Pointer To Pointer
- ☒ b) Indirect Base Class
- ☒ c) Polymorphism
- ☒ d) Null Pointer
- ☒ e) Memory Leak

2. Explain the concept of Inheritance in C++ with the help of example.

OR

Write a note on Early Binding and Late Binding in C++.

3. What is the use Data Files and write various classes available to handle files in C++.

OR

Explain the concept of pointer arithmetic with the help of example.

Department of Computer Science and Engineering
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OBJECT ORIENTED PROGRAMMING (Year -1, Semester - 2nd)

Max Marks = 15

Note: Section-A is compulsory. Attempt any two questions from Section-B

(5X1=5)

Section-A

- 1 (a) In which situation, this pointer is used?
(b) Can constructors be virtual? Give reason.
(c) How early binding is different from late binding?
(d) What are dangling pointers?
(e) Write the advantage of using function overloading.

(2X5=10)

Section-B

2. Discuss the various types of inheritance in C++ with the help of suitable examples.
3. Differentiate between function and class templates.
4. Write a program in C+ to illustrate the concept of various exception handling mechanisms.
5. Explain the following type conversions: basic type to class type and class to class type in C++.

Answer 1180
Department of Computer Science & Engineering, Punjabi University, Patiala

MST2

MM: 15

B.Tech-2nd Sem (CSE)

Time: 1 hour

OBJECT ORIENTED PROGRAMMING USING C++

Note: Q1 is Compulsory. Attempt any two questions from Q2, Q3 and Q4 each question carrying 5 marks.

(Q1) Explain the following (5 Marks)

- a/ Function Templates (1 Marks) b/ Pointer to an object (1 Marks)
c/ Array of pointers (1 Marks) d/ File opening modes (2 Marks)

Q2. Demonstrate the concept of hybrid inheritance by writing suitable examples. How ambiguity is resolved in inheritance?

Q3. What are virtual functions? How do they help in late binding? Give suitable example.

Q4. What is exception handling? Write down a sample program to demonstrate this.